

AI/ML Internship – Day 56 Notes & Tasks

Module 7: Word Embeddings (Word2Vec) – Understanding Semantic Meaning of Words

Part 1: Recap of Day 55

Yesterday we learned:

-  TF-IDF
-  Term Frequency (TF)
-  Inverse Document Frequency (IDF)
-  Importance of Words

Today we will learn:

-  Word Embeddings
-  Word2Vec

One of the most important concepts in Modern NLP.

Part 2: Problem with Bag of Words and TF-IDF

Suppose we have:

King
Queen
Man
Woman

BoW and TF-IDF see:

King $\neq$ Queen
Man $\neq$ Woman

They only count words.

They do not understand meaning.

Example

Sentence 1:

I love cars

Sentence 2:

I love automobiles

Humans understand:

Cars = Automobiles

But BoW treats them as completely different words.

Part 3: What is Word Embedding?

Definition

Word Embedding is a technique that converts words into numerical vectors while preserving their meaning and relationships.

Simple Meaning

Instead of:

Dog = Word

Cat = Word

We represent them as:

Dog = [0.25, 0.80, 0.45]

Cat = [0.28, 0.82, 0.40]

Notice:

Dog and Cat vectors are similar.

Because their meanings are similar.

Part 4: Why Word Embeddings?

Computers understand numbers.

But we want computers to understand:

☒ Meaning

☒ Similarity

✓ Relationships

Word Embeddings help achieve this.

📌 Part 5: What is Word2Vec?

Definition

Word2Vec is a Word Embedding technique developed by researchers at Google.

It learns word meanings by analyzing surrounding words.

Basic Idea

Tell me who your friends are, and I will tell you who you are.

Similarly:

Words appearing in similar contexts usually have similar meanings.

📌 Part 6: Example of Context Learning

Sentence:

The cat is sleeping on the bed

Sentence:

The dog is sleeping on the bed

Word2Vec notices:

cat

and

dog

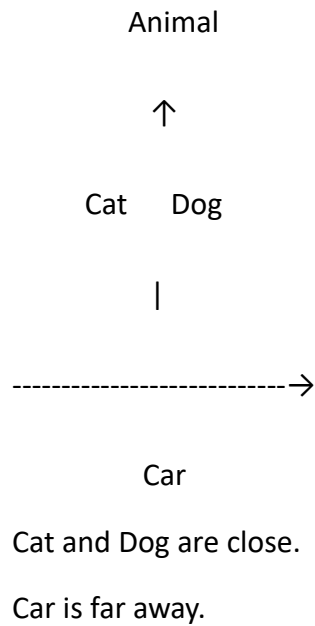
appear in similar positions.

Therefore:

Cat $\approx$ Dog

📌 Part 7: Word Embedding Visualization

Imagine a vector space:



Part 8: How Word2Vec Works

Word2Vec learns from millions of sentences.

Example:

The king rules the kingdom

The queen rules the kingdom

Word2Vec learns:

King $\approx$ Queen

because both appear in similar contexts.

Part 9: Famous Word2Vec Example

One of the most famous NLP examples:

King - Man + Woman = Queen

Why?

Because Word2Vec captures relationships.

Visual:

King

|

Male Royalty

|

Queen

|

Female Royalty

Part 10: Word Embedding Dimensions

Each word is represented by many numbers.

Example:

Dog

↓

[0.12, 0.56, 0.78, 0.34]

Real projects often use:

50 Dimensions

100 Dimensions

200 Dimensions

300 Dimensions

Part 11: Types of Word2Vec

1. CBOW

Continuous Bag of Words

Predicts:

Target Word

from surrounding words.

Example:

I ____ AI

Predict:

love

2. Skip-Gram

Predicts:

Surrounding Words

from target word.

Example:

Target:

love

Predict:

I

AI

📌 Part 12: CBOW vs Skip-Gram

CBOW	Skip-Gram
Predict center word	Predict surrounding words
Faster	Slower
Good for large datasets	Good for small datasets
Less accurate for rare words	Better for rare words

📌 Part 13: Word2Vec Using Python

Install:

pip install gensim

Import:

```
from gensim.models import Word2Vec
```

Part 14: Create Sample Dataset

```
sentences = [  
    ["i", "love", "ai"],  
    ["ai", "is", "amazing"],  
    ["i", "love", "python"],  
    ["python", "is", "powerful"]  
]
```

Part 15: Train Word2Vec Model

```
from gensim.models import Word2Vec
```

```
model = Word2Vec(  
    sentences,  
    vector_size=50,  
    window=3,  
    min_count=1  
)
```

Part 16: Get Word Vector

```
print(  
    model.wv["ai"]  
)
```

Output:

```
[0.24 0.18 0.75 ...]
```

Many numerical values.

Part 17: Find Similar Words

```
print(  
    model.wv.most_similar("ai")  
)
```

Example Output:

```
[  
    ('python', 0.82),  
    ('amazing', 0.70)  
]
```

Part 18: Real-World Example

Sentence:

I bought a car

Sentence:

I bought an automobile

BoW:

Different

Word2Vec:

Similar Meaning

Part 19: Applications of Word Embeddings

Chatbots

Understanding user intent.

Search Engines

Finding similar words.

Machine Translation

Language conversion.

Recommendation Systems

Content recommendation.

Sentiment Analysis

Better understanding of reviews.

Part 20: Why Word2Vec is Better than TF-IDF?

TF-IDF	Word2Vec
Counts importance	Understands meaning
No semantic relationship	Captures relationships
Sparse vectors	Dense vectors
Traditional NLP	Modern NLP

Part 21: Theory Questions

What is Word Embedding?

A numerical representation of words that captures meaning.

What is Word2Vec?

A technique used to generate word embeddings.

Why use Word2Vec?

To understand word similarity and context.

What are the two types of Word2Vec?

CBOW and Skip-Gram.

Give a famous Word2Vec example.

King - Man + Woman = Queen

Part 22: Interview Questions

Difference between TF-IDF and Word2Vec?

TF-IDF measures importance.

Word2Vec captures meaning.

What is CBOW?

Predicting a target word from surrounding words.

What is Skip-Gram?

Predicting surrounding words from a target word.

Why are embeddings important?

They help machines understand semantic relationships.

What library is commonly used for Word2Vec?

Gensim

Tasks for Day 56

Theory

1. Define Word Embedding.
 2. What is Word2Vec?
 3. Explain CBOW.
 4. Explain Skip-Gram.
 5. Why is Word2Vec better than TF-IDF?
-

Practical Task 1

Install:

```
pip install gensim
```

Practical Task 2

Create a small dataset with 5 sentences.

Practical Task 3

Train a Word2Vec model.

Practical Task 4

Print word vectors.

Practical Task 5

Find similar words using:

```
most_similar()
```